

Operator Discovery Report: candidate_pruner_cluster_grid_stagnation_adaptive

- Operator type: candidate_pruner.
- Host scaffold: nearest_neighbor_2opt.
- Core idea: Deterministic adaptive candidate pruning for local moves: keeps a moderate candidate list by default, expands it under stagnation and for clustered instances (where useful exchanges are more numerous), and tightens it for
- Novelty classification: descriptor_adaptive_pruner.
- Rediscovery signature: candidate_pruner:18:2:-4:6.
- Surviving candidate: False.
- Problem class where it helps: none confirmed.

Pseudocode

- Start from a base candidate-list size.
- Expand or shrink the list using instance descriptors such as size, trap score, grid-likeness, clustering, and current stagnation.
- Run the downstream local search only on the resulting pruned neighborhood.

Complexity

- Non-empty lines: 12.
- Complexity score: 0.44.
- Fingerprint: 2e617516c67ae59c.

Why It Should Help

- The operator adapts neighborhood size to instance structure, which can trade off search breadth and runtime in a reusable way.

Failure Cases

- Does not transplant cleanly into clustered_local_search.
- No validation family shows a clear improvement over the host baseline.
- Disabling the operator does not hurt, so the ablation does not support a real operator effect.

Transplant Results

- nearest_neighbor_2opt gap delta 0.000609, runtime delta 74.223843, distance-eval delta 1493.0.
- cheapest_insertion_2opt gap delta 0.0, runtime delta 1165.137014, distance-eval delta 965.142857.
- random_restart_2opt gap delta 0.001067, runtime delta 70.841629, distance-eval delta 4993.428571.
- sparse_three_opt gap delta -0.000253, runtime delta 37.929086, distance-eval delta 3605.857143.
- clustered_local_search gap delta 0.013098, runtime delta 245.002514, distance-eval delta 10760.0.

Instance-Family Results

- heldout_tsplib operator gap 0.235667 vs baseline 0.235058.
- clustered_tsp operator gap 0.110853 vs baseline 0.110853.
- elongated_corridor_tsp operator gap 0.003204 vs baseline 0.003204.
- grid_like_tsp operator gap 0.106696 vs baseline 0.106696.
- nearest_neighbor_trap_tsp operator gap 0.0 vs baseline 0.0.
- two_cluster_bottleneck_tsp operator gap 0.109057 vs baseline 0.109057.
- uniform_euclidean operator gap 0.0 vs baseline 0.0.

Pareto Results

- Gap delta: 0.000224.
- Runtime inflation: 0.179435.
- Distance-evaluation inflation: 0.156445.
- Same-gap-faster flag: False.
- Better-gap-same-runtime flag: False.

Ablation Results

- disabled mean gap 0.121317 and delta vs full operator -0.000224.
- constant_pruner mean gap 0.121542 and delta vs full operator 0.0.
- wide_pruner mean gap 0.121542 and delta vs full operator 0.0.